

Milind V Gunjal

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Employment

Adjunct Assistant Professor (Tallahassee State College) Instructor of Record	2025-Present
Administrative Teaching Assistant (Florida State University) - Instructor of Record - Course development - Coordinator of volunteers for the AMS sectional meeting - Office assignments for teaching assistants - Computer lab resource management - Qualifiers proctoring	2023-2025
Teaching Assistant (Florida State University) - Instructor of Record - Lab Instructor	2019-2023 2020-Present 2019-2020

Education

PhD at Florida State University	2019-2025
MS at Florida State University (CGPA: 3.91/4)	2019-2021
BS-MS at IISER-Kolkata (CGPA: 8.56/10, Valedictorian)	2014-2019

Areas of interest

Mathematics • Algebraic topology, Category theory, Homotopy theory, K-theory.

Fellowships & awards

Dwight B. Goodner Mathematics Fellowship	2024
Distinguished Teaching Assistant	2023
Kishore Vaigyanik Protsahan Yojana (KVPY) Fellow	2014-2019
Indian Institute of Technology Joint Entrance Exam Advanced Qualified	2014

Research projects undertaken

Cocycles with values in a Picard category from an SM 2-Cat (PhD thesis)	10/2023- 04/2025
Advisor: Dr. Ettore Aldrovandi, FSU	
Comparison of Segal and Waldhausen K-theory via MacLane cubes	08/2023- 08/2024
Joint with Dr. Ettore Aldrovandi, Dr. Brandon Doherty, Arash Karimi, FSU	
K-theory of a Waldhausen Category	10/2020- 04/2025
Joint with Dr. Ettore Aldrovandi, FSU	
Vector bundles from algebraic viewpoint (MS thesis)	08/2018- 05/2019
Supervisor: Dr. Somnath Basu, IISER-K	
Representation theory of $GL(2, F_q)$	08/2018- 12/2018
Supervisor: Dr. Utsav Choudhury, ISI-Kolkata	
Study of simplicial and singular homology	05/2018- 07/2018
Supervisor: Dr. Priyavrat Deshpande, CMI, Chennai	
Hilbert Nullstellensatz	01/2018- 05/2018
Supervisor: Dr. Somnath Basu, IISER-K	
Study of modules of ring of continuous functions	08/2017- 12/2017
Supervisor: Dr. Somnath Basu, IISER-K	
Study of evolutionary game theory	05/2017- 07/2017
Supervisor: Dr. Supratim Sengupta, IISER-K	
Study of elementary number theory and cryptography	05/2016- 07/2016
Supervisor: Dr. S A Katre, SPPU, Pune	

Seminars and Conference Talks

Introduction to stable homotopy theory (Advanced Seminar in Algebra at FSU) (Notes)	04/2024
AMS Sectional Meeting at FSU (Slides)	03/2024
Lloyd Roeling Conference at UL (Slides)	03/2024
Cohomology with values in a Picard Category (Advanced Seminar in Algebra at FSU) (Slides)	12/2023
Stabilization of 2-Crossed Modules (BUGCAT'23) (Slides)	11/2023
K-theory of a Waldhausen category (Kan Seminar) (Slides)	11/2023
Homotopy theory of Model Categories (Advanced Seminar in Algebra at FSU) (Notes)	04/2023
Stabilization of 2-Crossed Modules (Advanced Seminar in Algebra at FSU) (Slides)	04/2022
2-type of the K-theory of a Waldhausen category (FSU) (Slides)	12/2021

Seminars and Conferences Attended

Algebraic Structures in Topology, San Juan, PR	06/2024
Graduate Student Topology and Geometry Conference 2024, MSU	04/2024
AMS Sectional Meeting, FSU	03/2024
Lloyd Roeling Conference, UL	03/2024

Binghamton University Graduate Combinatorics Algebra Topology Conference	11/2023
Online workshop on $(\infty, 2)$ -Categories	10/2023-11/2023
Kan Seminar	09/2023-12/2023
eCHT Research Seminar	09/2023-11/2023
Summer school on Scissors Congruence, Algebraic K-Theory, and Trace Methods, IU	06/2023
Mid-Atlantic Topology Conference, UPenn	04/2023
Advance Seminar in Topology and Geometry, FSU	08/2019 - Present
Advanced Seminar in Algebra, FSU	08/2019 - Present
Seminar series on Moduli spaces, IISER-K	11/2018
FACETS, IMSc, Chennai	07/2018
Young Topologists' Meet, CMI, Chennai	07/2018
Madhava Mathematics Competition Nurture Camp, CMI, Chennai	06/2018
Learning seminar on Morse Theory, IISER-K	01/2018 - 04/2018

Teaching

Mathematical Thinking (MGF1130) (Instructor of Record at TSC)	Spring 2026
College Algebra (MAC1105) (Instructor of Record at TSC, 3 sections)	Fall 2025
Calculus III (MAC2313) (Instructor of Record at FSU)	Fall 2024
Calculus II (MAC2312) (Instructor of Record at FSU)	Fall 2023
Calculus II (MAC2312) (Instructor of Record at FSU)	Spring 2023
Discrete Mathematics I (MAD2104) (Recitation Instructor at FSU)	Fall 2022
Calculus I (MAC2311) (Instructor of Record at FSU)	Summer 2022
Calculus I (MAC2311) (Instructor of Record at FSU)	Spring 2022
Calculus I (MAC2311) (Recitation Instructor at FSU)	Fall 2021
Precalculus Algebra (MAC1140) (Instructor of Record at FSU)	Spring 2021
Precalculus Algebra (MAC1140) (Instructor of Record at FSU)	Fall 2020

Advanced Courses Undertaken

- **Stable Homotopy Theory** ref. Cary Malkiewich - Spectra and stable homotopy theory (eCHT, WSU)
- **Symplectic Geometry** ref. Ana Cannas da Silva - Lecture Notes (FSU)
- **Differential Geometry of Bundles** ref. Sharpe - Differential Geometry (FSU)
- **Algebraic Geometry** ref. David Eisenbud and Joe Harris - 3264, and Robin Hartshorne - Algebraic Geometry (FSU)
- **Homological Algebra** ref. C. Weibel - An Introduction to Homological Algebra (FSU)
- **Algebraic Geometry by way of Scheme Theory** ref. Ravi Vakil - The Rising Sea: Foundations of Algebraic Geometry (FSU)

- **Geometric Structures on Manifolds** (FSU)
- **Logic, Type Theory, and the Mechanization of Mathematics** (FSU)
- **Differential Topology II** ref. John Lee - Introduction to Smooth Manifolds (FSU)